Ideation Phase

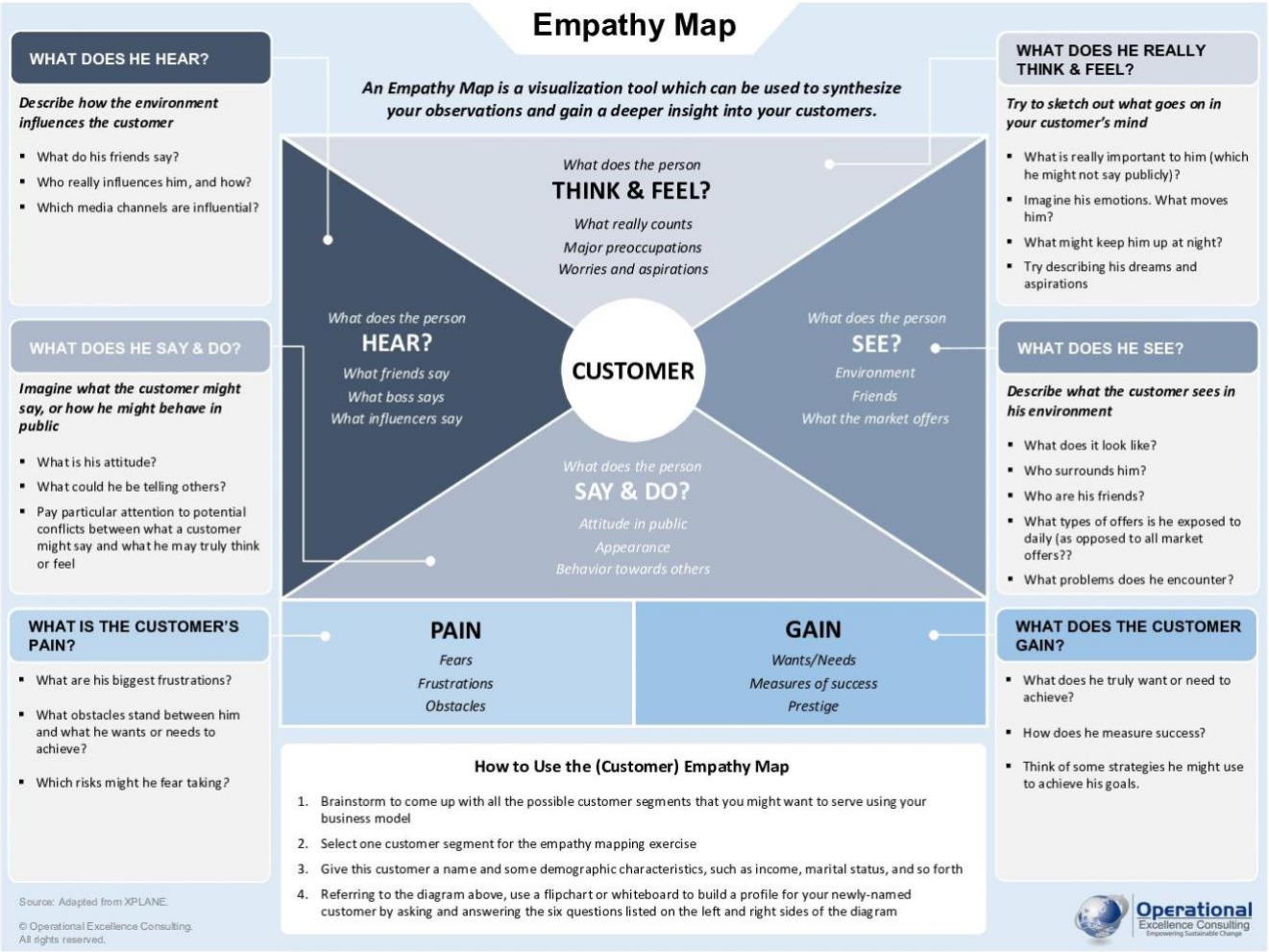
Empathize & Discover

Date	10 February 2026
Team ID	LTVIP2026TMIDS3483
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Empathy Map Canvas:

The empathy map for Plugging into the future: An exploration of electricity consumption patterns project focuses on electricity consumers such as households and small business owners who are directly affected by changing energy consumption patterns. These users see their monthly electricity bills, smart meter readings, and increasing usage of electrical appliances in daily life. They hear news about rising electricity tariffs, energy conservation campaigns, and advice from others about reducing power consumption. They think and feel concerned about high electricity costs, confused about sudden bill increases, and motivated to manage their usage more efficiently. They often say things like, “My electricity bill is too high,” and take actions such as turning off unused appliances, using energy-efficient devices, or monitoring their consumption more carefully. However, they experience pains like lack of clear understanding of usage trends, unexpected spikes in bills, and difficulty identifying peak consumption periods. Their gains include wanting accurate insights, clear visual reports, cost-saving suggestions, and better control over electricity usage. This project addresses these needs by analyzing electricity consumption data and presenting meaningful visualizations, helping users make informed and sustainable energy decisions.

Example:


Empathy Map
An Empathy Map is a visualization tool which can be used to synthesize your observations and gain a deeper insight into your customers.
WHAT DOES HE HEAR?
Describe how the environment influences the customer
What do his friends say?
Who really influences him, and how?
Which media channels are influential?
WHAT DOES HE SAY & DO?
Imagine what the customer might say, or how he might behave in public
What is his attitude?
What could he be telling others?
Pay particular attention to potential conflicts between what a customer might say and what he may truly think or feel
WHAT IS THE CUSTOMER'S PAIN?
What are his biggest frustrations?
What obstacles stand between him and what he wants or needs to achieve?
Which risks might he fear taking?
What does the person
THINK & FEEL?
What really counts
Major preoccupations
Worries and aspirations
What does the person
HEAR?
What friends say
What boss says
What influencers say
CUSTOMER
What does the person
SEE?
Environment
Friends
What the market offers
What does the person
SAY & DO?
Attitude in public
Appearance
Behavior towards others
PAIN
Fears
Frustrations
Obstacles
GAIN
Wants/Needs
Measures of success
Prestige
WHAT DOES HE REALLY THINK & FEEL?
Try to sketch out what goes on in your customer's mind
What is really important to him (which he might not say publicly)?
Imagine his emotions. What moves him?
What might keep him up at night?
Try describing his dreams and aspirations
WHAT DOES HE SEE?
Describe what the customer sees in his environment
What does it look like?
Who surrounds him?
Who are his friends?
What types of offers is he exposed to daily (as opposed to all market offers??
What problems does he encounter?
WHAT DOES THE CUSTOMER GAIN?
What does he truly want or need to achieve?
How does he measure success?
Think of some strategies he might use to achieve his goals.
How to Use the (Customer) Empathy Map
1. Brainstorm to come up with all the possible customer segments that you might want to serve using your business model
2. Select one customer segment for the empathy mapping exercise
3. Give this customer a name and some demographic characteristics, such as income, marital status, and so forth
4. Referring to the diagram above, use a flipchart or whiteboard to build a profile for your newly-named customer by asking and answering the six questions listed on the left and right sides of the diagram
Source: Adapted from XPLANE.
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Example:

Plugging into the future:An exploration of electricity consumption patterns

Plugging into the Future:

An Exploration of Electricity Consumption Patterns

THINKS

- How is electricity demand growing yearly?
- Which region consumes the most power?
- How can peak load be reduced?
- What are seasonal usage trends?
- How to improve renewable adoption?

SAYS

- We need better visualization tools.
- Compare region-wise consumption.
- Show monthly and yearly growth clearly.
- We must plan for future demand.

FEELS

- Overwhelmed by large datasets.
- Concerned about energy shortages
- Responsible for public energy planning.
- Pressured to improve sustainability.
- Worried about CO_2 emissions

Energy Policy Analyst / Electricity Board Planner

(Goal: To analyze electricity consumption patterns and make data-driven energy decisions)

SAYS

- We need better visualization tools.
- Compare region-wise consumption.
- Show monthly and yearly growth clearly.
- We must plan for future demand.

DOES

- Collects electricity data
- Uses Tableau dashboards
- Compares states & sectors
- Tracks seasonal peaks
- Supports smart energy planning

PAIN POINTS

- No centralized dashboard
- Difficult to compare multiple dimensions
- Hard to identify peak consumption quickly
- Manual data analysis is time-consuming

GAINS

- Interactive dashboards
- Clear seasonal and regional comparison
- Easy identification of peak months
- Supports smart energy planning

Empathy Map